Agricultural Risk Management & Weather Derivative Replicating Platform

ECE 473 / COS 473 — Milestone II: Contract Design & Platform Architecture

Maya, Evan, Jack

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Executive Summary

Showcase: the PRF Companion. A working web app that keeps a rancher's USDA rainfall insurance and uses Kalshi contracts to cover the months it leaves open. It tests each month at the rancher's location, recommends only the months that track, and builds the order. Fit on 1951–2000 and scored on 2001–2025, the companion has a median hedge effectiveness of 0.61, against 0.23 for hedging the PRF intervals themselves. It is recommended for 9 of our 11 farms (Section 4).

Milestone I measured how far today's Kalshi monthly-rain contracts are from USDA's Pasture, Rangeland and Forage (PRF) rainfall-index insurance. Milestone II builds the tools the proposal promised and uses them to answer two questions: *what is the best hedge a farmer can buy*, and *what would have to change for it to be good*. All results use the 11 representative farms at our 7 PRF study sites, their 2026 PRF elections, and the NCEI-based PRF index that PRF pays on from crop year 2027, over 1951–2025. Hedge effectiveness (HE) is the share of the year-to-year variance of the farm's PRF payout that a hedge removes. We report it *out of sample*: the hedge is refit 75 times, leaving out one year each time, and scored on the year it did not see.

1. **Optimizer (Week 4, Jack).** A convex program that picks the lowest-error portfolio of Kalshi contracts for any budget, with prices, fees, capital cost and order-book depth. The cheapest hedge with 90% of the best fit costs 22% less than the lowest-error hedge and loses nothing out of sample.
2. **Error breakdown (Week 4, Maya).** Out of sample, HE falls from 1.00 to 0.17: geography -0.35, settlement source -0.13, monthly contracts -0.19, whole-inch strikes ≈ 0, order-book depth -0.16, and a budget equal to the PRF premium -0.01.
3. **Contract gaps and proposals (Week 4, Evan).** Only 3 of 36 insured windows replicate well today; summer convective and monsoon rain fails everywhere. Two-month contracts raise pooled HE from 0.34 to 0.48; binaries on the PRF grid index itself reach 0.98, but that is PRF without the subsidy.
4. **Platform (Week 5, Jack and Maya).** A working web app: a farmer clicks a location, picks any months, and gets an order ticket priced from live Kalshi order books where markets are listed, a price-vs-fit frontier, a 75-year backtest, and a comparison with PRF.
5. **Complementing PRF (refinements promised in our Milestone I update).** The best niche is the months PRF leaves uncovered: single-month cover for November–February copies about half of the payout swings (median HE 0.52; December 0.68). A top-up above PRF's 90% cap copies little (0.14). Excess-rain cover needs two-month contracts (0.22 to 0.40). Backtests on the CPC index USDA actually paid tell the same story.

6. **App test (Week 5, Evan).** Typing the 11 farms into the app reproduces the analysis exactly. The recommended hedges cost 1.71× the farmers' PRF premiums (median) and copy a bit over a third of PRF's payout swings out of sample (median HE 0.38). Across 150 random grids the median HE is 0.47 within 25 miles of a station and 0.15 beyond 100 miles; 69% of sampled grids score below 0.3.
7. **Showcase: PRF Companion.** Built on these results; see Section 4.

1 Designing Better Options (Week 4)

1.1 Lowest-error, lowest-cost portfolios (Jack)

Problem. In year t the farmer wants y_t dollars (the PRF indemnity). A *leg* is a NO contract on "station rain in month m above K inches", paying \$1 if rain $\le K$. Each leg can be bought in *tranches*: price levels of an order book with sizes, or an unlimited quantity at a model price. With x_j contracts in tranche j, all-in unit cost c_j and hedge payout h_t,

$$\min_x \ \frac{1}{n}\sum_t (y_t - h_t)^2 \quad \text{s.t.} \quad \sum_j c_j x_j \le B, \qquad 0 \le x_j \le \text{cap}_j,$$

where $c_j = (p_j + 0.07\,p_j(1-p_j))(1 + r\,d/365)$ adds Kalshi's taker fee and the cost of prepaying cash ($r = 4.05\%$, $d = 31$ days). We solve the penalized form with L-BFGS-B and find the budget form by bisection on the penalty. Sweeping B traces the *efficient frontier*. With $B = \infty$ and no caps it reproduces Milestone I's least-squares hedge exactly.

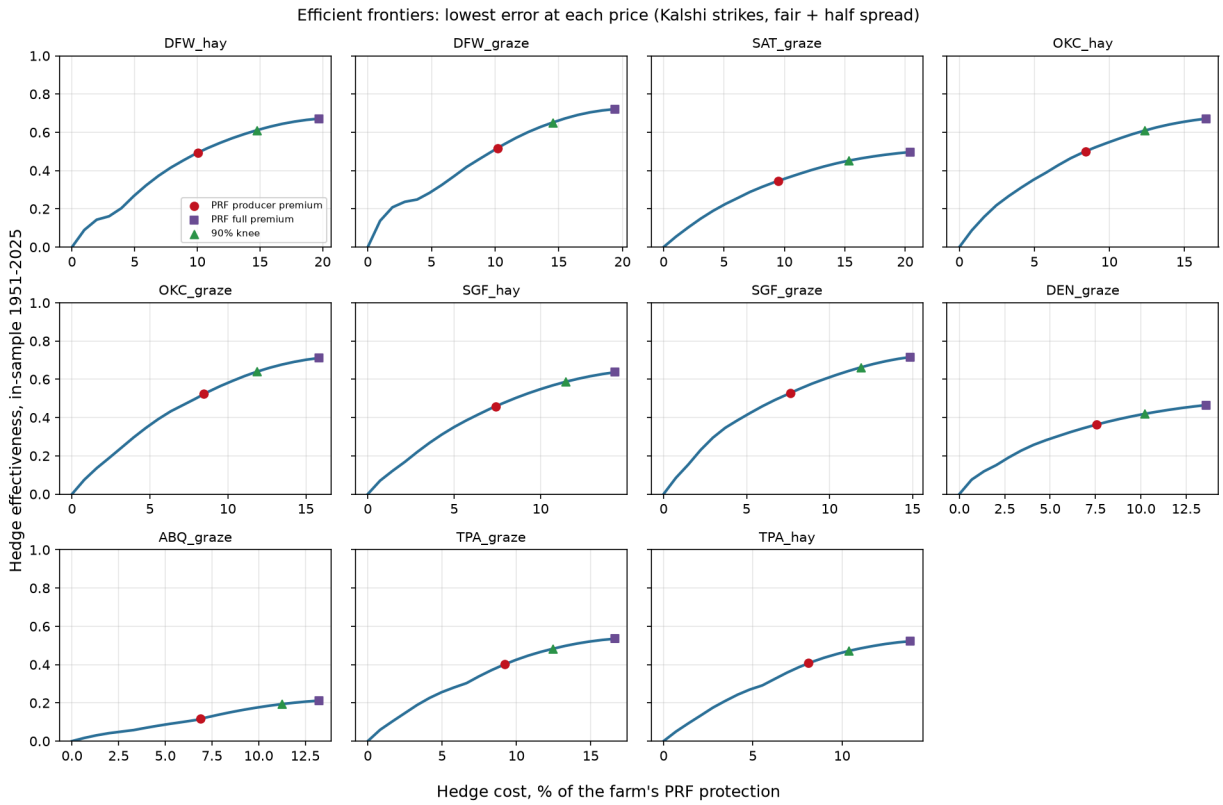


Figure 1: Efficient frontiers for the 11 farms (in-sample HE vs. cost as % of PRF protection). Red: budget equal to the farm's PRF producer premium. Green: the "90% knee". Purple: full PRF premium.

Point on the frontier (11 farms)	Cost / yr	$ per $1 expected	Median HE in	Median HE out
Lowest error (no budget)	$104,970	1.10	0.64	0.37
90% knee	$81,516	1.11	0.59	0.36
Budget = PRF producer premium	$55,992	1.12	0.46	0.30
PRF itself: producer / full premium	$55,994 / $117,053	0.61 / 1.28	1	1

Table 1: Four points on the frontier, priced at fair value plus half a 4.5-cent spread.

Findings. (i) In-sample fit overstates real performance by about 40% (median 0.64 vs 0.37), so the platform always reports the out-of-sample number. (ii) The knee costs 22% less with no loss out of sample. A budget acts as regularization, and it beats the lowest-error hedge out of sample for six farms. (iii) The full PRF premium never binds: without the subsidy Kalshi is cheaper than PRF's list price, but it buys a much worse match.

1.2 Decomposing the remaining error (Maya)

Starting from a perfect contract, we add one real-world restriction at a time and refit the best hedge: the PRF formula on the *station's* grid (geography); on the station *gauge* (settlement source); on each month separately with 0.25-inch strikes (monthly contracts); at Kalshi's listed whole-inch strikes (binning); at most the NO depth within 10 ¢ of fair value on the post-rewards November 2026 book (liquidity); and total cost at most the PRF producer premium (capital).

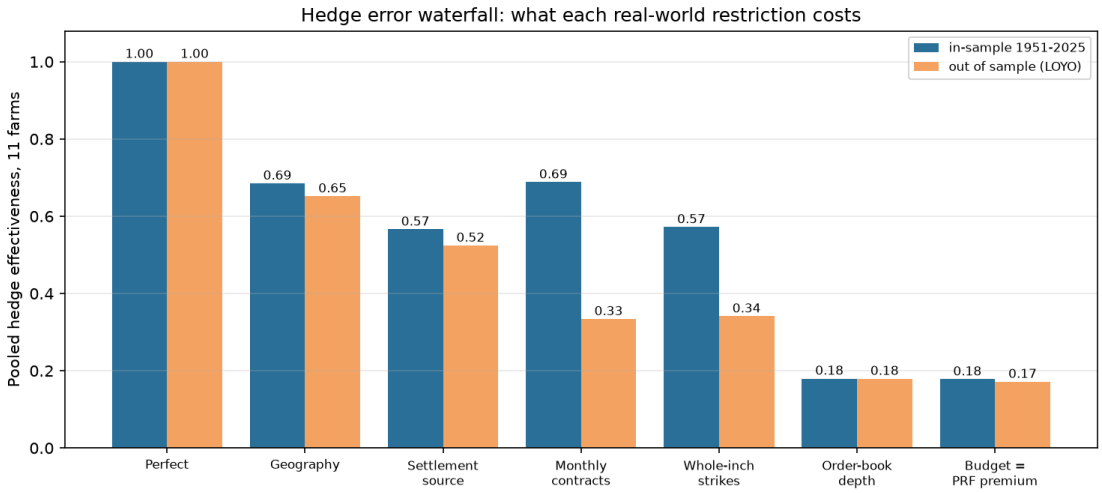


Figure 2: Pooled HE after each restriction, 11 farms, 1951–2025.

Findings. Geography is the largest loss (−0.35; the largest drop in 23 of 36 insured windows). Settlement source costs a steady −0.13. Splitting PRF's two-month interval into monthly contracts costs −0.19. Whole-inch binning costs nothing out of sample: finer strikes look 0.12 better in sample, but that gain is overfitting. Order-book depth costs −0.16 (the largest loss in sample, −0.39). Once depth binds, the capital constraint barely matters. Two thirds of the total loss (0.48 of 0.83) comes from paying on the wrong rain, which no strike set or optimizer can fix.

1.3 Exposures that cannot be replicated, and what would fix them (Evan)

Contract set	Pooled HE out	Farms with HE ≥ 0.6	\$ per \$1 expected
Today: monthly, whole-inch strikes	0.34	0 of 11	1.10
P1 monthly, half-inch strikes	0.34	0 of 11	1.12
P2 two-month total, 1-inch strikes	0.48	2 of 11	1.13
P3 two-month total, half-inch strikes	0.47	2 of 11	1.14
P4 binaries on the farm's grid index, 10-point steps	0.98	11 of 11	1.20
P5 same, 5-point steps	0.99	11 of 11	1.20

Table 2: Proposed contracts (lowest-error hedge, no depth or budget limit).

Exposures that fail. Only 3 of 36 insured windows reach HE 0.6 today (OKC spring and SGF fall); 21 are partial and 12 fail. The failures are summer convective or monsoon rain: every Albuquerque window, Denver Jun–Sep, Tampa May–Jun, Springfield May–Jun, Dallas Aug–Sep, San Antonio Sep–Oct.

What would fix them. (1) *Two-month contracts* (P2) need no new data source and improve 35 of 36 windows. They are the first thing to propose to the exchange. (2) *Grid-index binaries* (P4) recover almost everything, but they are PRF without the subsidy and need NOAA gridded data as a settlement source. (3) *Market mechanisms*: the busiest leg needs 5–28× today's sensible depth, and only 16% of the contracts could be filled. Hedging needs ladders listed 6–12 months ahead (today about 30 days), a designated market maker on the drought side (rewarded depth fell 94% when rewards ended), and pooled orders through cooperatives or agents. Matching PRF's net benefit would need a subsidy of about $44,600 a year for the 11 farms.

2 Building the App (Week 5)

2.1 Live-price backend (Jack)

One function, `platform.quote`, serves the web app and the tests. It (1) maps the location to its PRF grid and ranks the Kalshi stations within 25 miles of the nearest by out-of-sample fit; (2) builds the PRF-style target for any chosen months; (3) prices each contract from Kalshi's public order-book API (cached 60 s, with a saved-snapshot fallback), skipping levels more than 10 ¢ above fair value; months not yet listed are priced at fair value plus half a typical spread and flagged; (4) runs the optimizer for the frontier and the recommended hedge; and (5) returns an order ticket with whole contracts, limit prices and fees. Two rules came from live testing. Coverage starts with the next month that has not begun, because a live test bought an in-progress November at 99 ¢ when its price already reflected the rain so far. The nearest station is not always the best: St. Petersburg is nearer to the Tampa farms than Tampa, but has only 17 years of record.

2.2 Farmer screen (Maya)

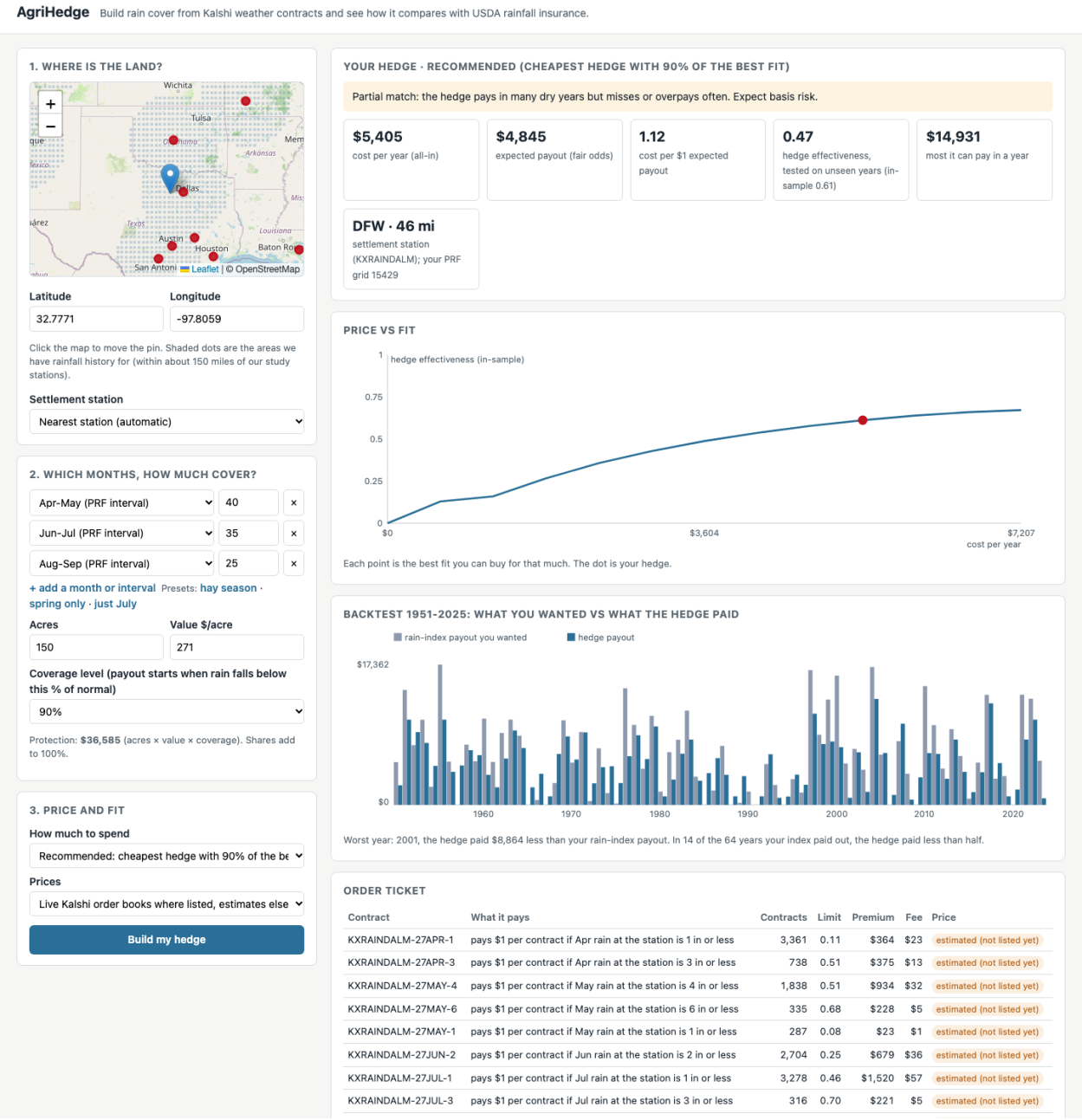


Figure 3: The farmer screen: location, months and cover on the left; verdict, price-vs-fit frontier, backtest, order ticket and PRF comparison on the right.

A farmer clicks the map (2,036 PRF grids around the 8 study stations are served), adds any PRF interval, single month or three-month season with a share of protection, enters acres, value per acre and coverage level, and chooses "recommended", "best fit" or a budget. The answer starts with a plain verdict (good, partial or poor match, from out-of-sample HE) and shows the live market depth at the station.

2.3 Testing the app against the USDA benchmark (Evan)

Each farm typed its own location, PRF intervals, coverage and protection into the app (station chosen automatically, estimated prices). Then a standard hay-season request was run at 150 random grids.

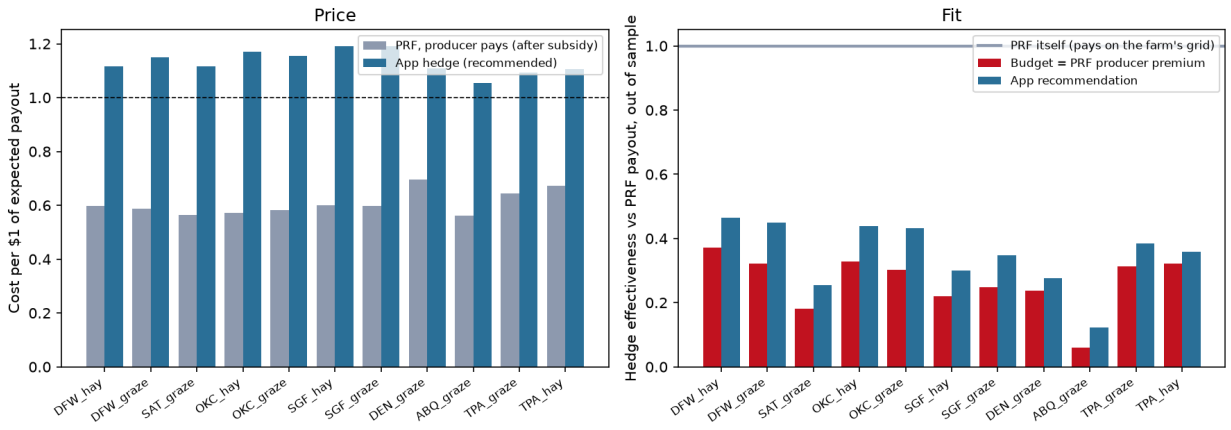


Figure 4: Left: cost per $1 of expected payout, PRF (producer share) vs. the app's recommended hedge. Right: out-of-sample HE against the PRF payout.

Measure (11 farms)	Value
PRF producer premiums / app recommended hedges	$55,994 / $95,182 per year
Hedge cost ÷ PRF producer premium, per farm	1.56–1.97 (median 1.71)
Cost per $1 expected payout: PRF producer / app hedge	$0.56–0.70 / $1.05–1.18
Median HE out of sample: recommended / PRF-premium budget / lowest error	0.38 / 0.30 / 0.37
Years the hedge paid < half of PRF, of years PRF paid	2–11 of 50–68 per farm
Farm-years the hedge paid but PRF did not	115
Worst single-year shortfall	$3,588 (SGF_graze) to $50,964 (ABQ_graze)
App matches doc analysis (station, grid, cost)	11 of 11; $0.00 cost gap
App PRF premium estimate vs. real 2026 premium	median +3% (−13% to +10%)

Table 3: The app against the USDA PRF benchmark.

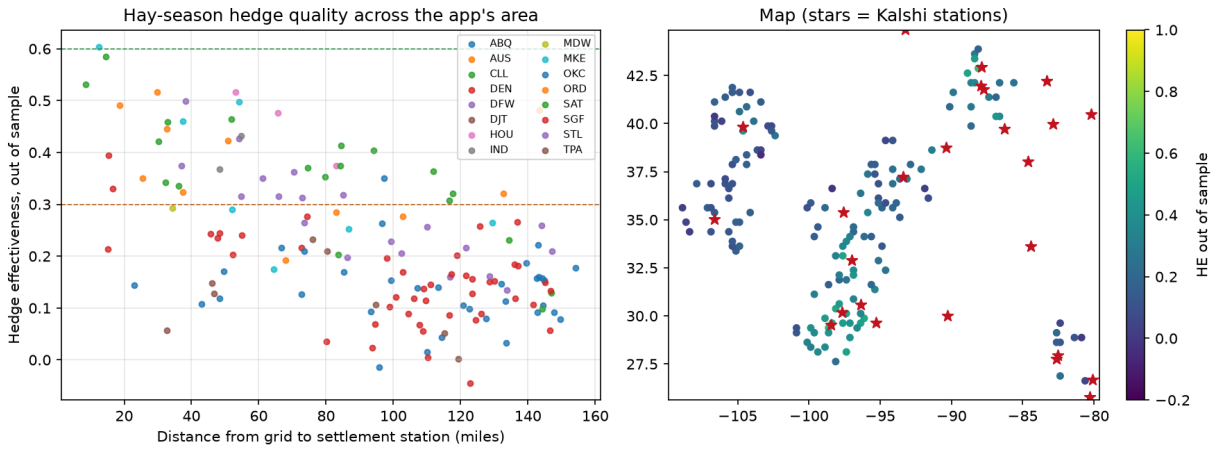


Figure 5: Hay-season hedge quality at 150 random grids: out-of-sample HE vs. distance to the settlement station (left) and on a map (right).

Findings. The app is correct, and its answer is sobering. Hedging PRF's own intervals costs about 1.7× what PRF costs the farmer and copies a bit over a third of PRF's swings. In 115 farm-years it paid when PRF did not, and those false alarms are what make it expensive. Distance decides who can use it: median HE is 0.47 within 25 miles, 0.35 at 25–50, and 0.15 beyond 100 miles. (The app now recommends the cheapest hedge with 95% of the best fit, which did best on unseen years across 44 cases.) With only 7.4% of US PRF grids within 50 miles of a Kalshi rain station (Milestone I), a useful hedge reaches a small minority of PRF land. On 2026-11-05 no month beyond the one in progress was listed, so every order for a real season was correctly flagged as estimated.

3 Complementing PRF (refinements from our Milestone I update)

Our Milestone I update said we would complement PRF rather than replace it, by targeting cover above PRF's 90% cap, uncovered months and too much rain. It also said we would look beyond binary payouts and backtest on the index USDA actually paid (CPC), with NCEI for forward results.

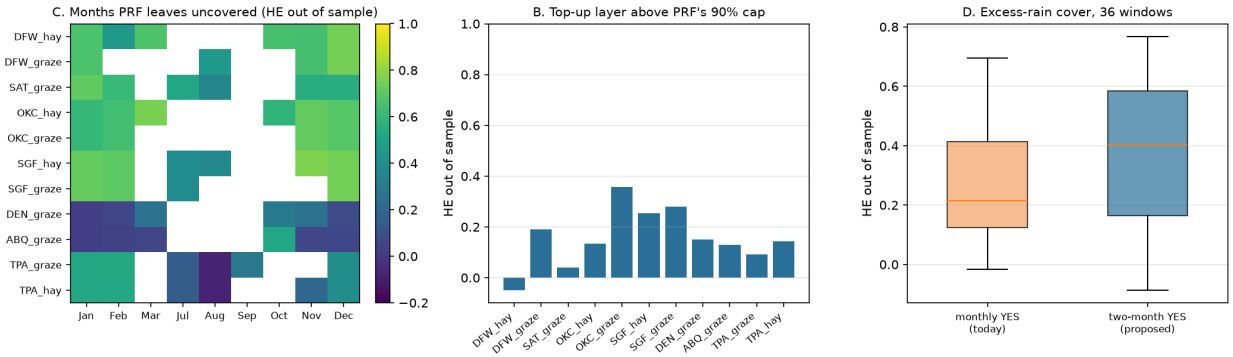


Figure 6: Left: out-of-sample HE of single-month cover for months outside each farm's PRF intervals. Middle: a top-up layer above PRF's 90% cap. Right: excess-rain cover with monthly vs. two-month YES contracts.

Gap PRF leaves	Median HE in	Median HE out of sample
Uncovered months, single-month cover at 90% (60 farm-months)	–	0.52 (26 of 60 ≥ 0.6)
Top-up layer above PRF's 90% cap (11 farms)	0.40	0.14
Excess rain (> 130% of normal), today's monthly YES (36 windows)	0.47	0.22
Excess rain, proposed two-month YES (36 windows)	0.70	0.40
PRF itself, CPC as paid vs. NCEI, 1969–2025 (lowest error / knee)	–	0.20 vs. 0.28 / 0.26 vs. 0.27

Table 4: Where event contracts can complement PRF.

Findings. Cool-season frontal rain spreads evenly over a region, so the airport gauge tracks the farm in the months PRF leaves open (December 0.68, November 0.65, January 0.60). That is the strongest use of today's contracts we found. The value is mainly for winter pasture, stock water and winter wheat, because warm-season forage responds little to winter rain. A top-up above 90% pays a little in nearly every year and copies poorly. Excess-rain cover works only with two-month contracts, which strengthens the case for them. On payouts beyond binaries: contracts paying the PRF formula linearly on the station gauge or station grid index reach 0.52–0.65 (Section 1.2). Grid-index binaries reach 0.98. So what matters is *what* a contract settles on, not the payout shape. Backtesting on the CPC index USDA actually paid gives slightly lower out-of-sample HE than NCEI on the same years. The 90% knee closes most of that gap. Temperature risk ("if time permits") is left for Milestone III.

4 Showcase: the PRF Companion

Sections 1–3 point to one product. Kalshi rain contracts are a poor *substitute* for PRF, but a good *complement* in the months PRF leaves open. The app's PRF Companion tab does exactly that:

1. The rancher enters a location and the PRF intervals they insure.
2. Each month PRF leaves open is scored by out-of-sample HE.
3. Months at 0.5 or above are recommended and combined into one Kalshi order.
4. A 1951–2025 backtest shows the companion's payouts next to what PRF paid.

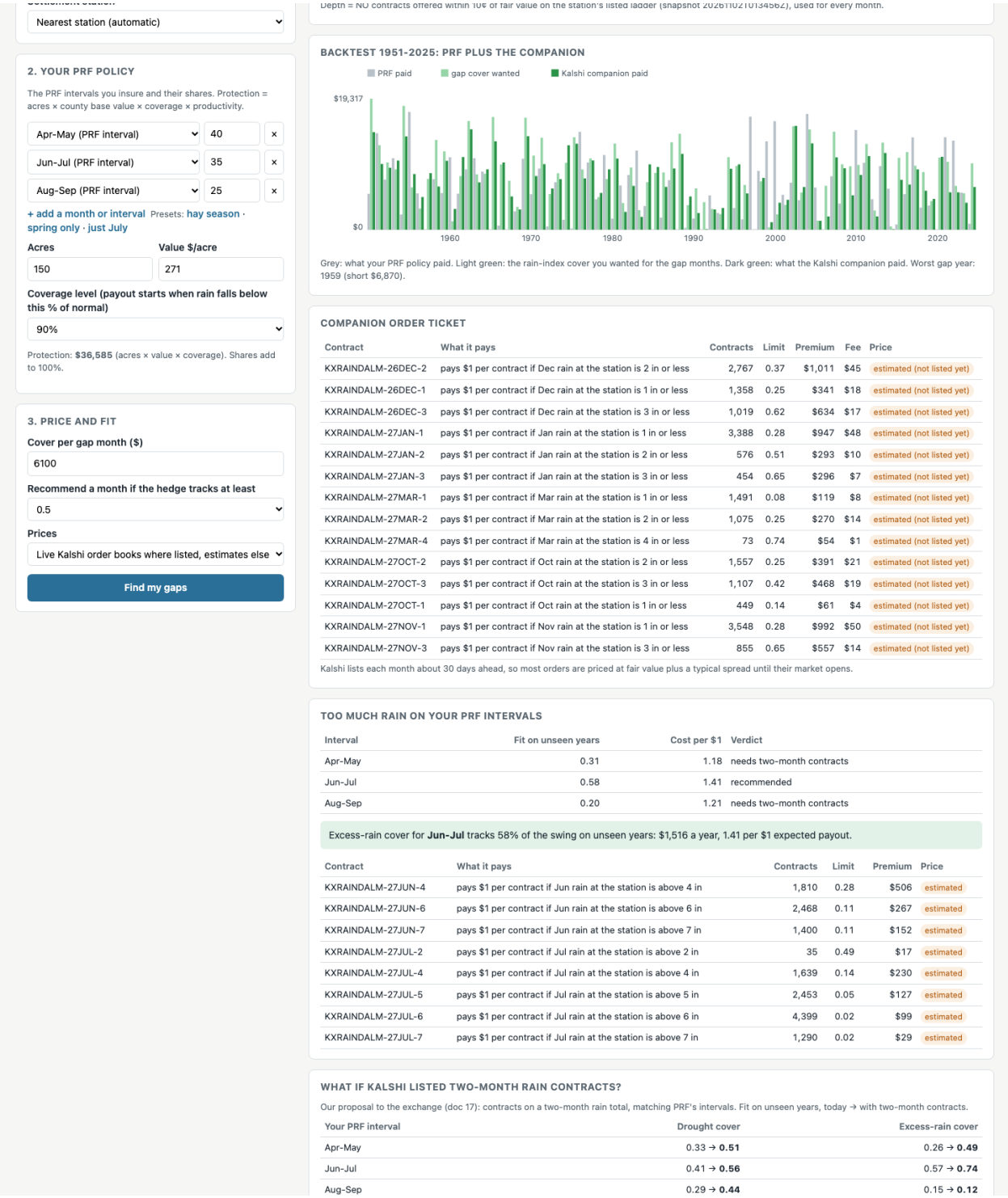


Figure 7: The PRF Companion for the Dallas hay farm (PRF on Apr–Sep). The month strip shows the recommended gap months (green), with their out-of-sample HE. Below it: the package, the backtest of PRF plus the companion, and the order ticket.

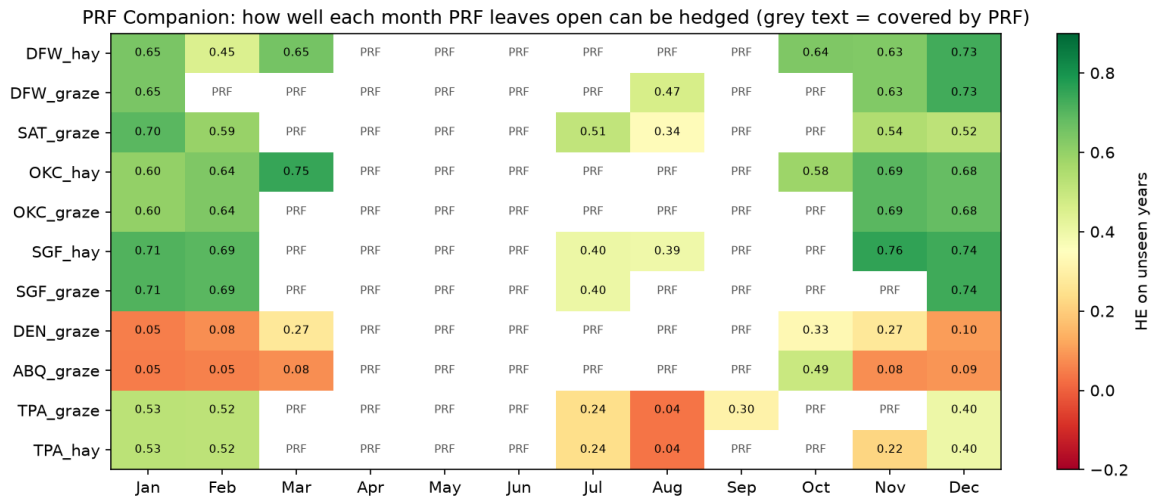


Figure 8: Out-of-sample HE of a Kalshi hedge for every month each farm's PRF policy leaves open.

Measure	PRF Companion	Hedging PRF's own intervals
Farms where it is recommended	9 of 11	–
Median HE out of sample (leave one year out)	0.66	0.38
Holdout: months chosen and fit on 1951–2000, scored on 2001–2025	0.61	0.23
Random grids with a useful result (60 grids / 150 grids)	60% with ≥ 1 month	1% with HE ≥ 0.6
Cost per \$1 of expected payout (median)	\$1.10	\$1.12

Table 5: The companion against hedging PRF itself.

Findings. December, November and January carry the companion: 7 of 11, 6 of 9 and 9 of 11 of those farm-months are recommended. Summer months almost never are. Picking months and fitting on 1951–2000 only, the companion still scores 0.61 on 2001–2025, so the result is not an artifact of choosing on the test years. Combining several nearby stations does not help, because Kalshi stations are too sparse; only the Tampa farms gain (0.37 to 0.43). Caveats: winter rain matters less for warm-season forage, so the value is greatest for winter pasture, winter wheat and stock water. Each month is listed only about 30 days ahead, so orders are placed month by month.

What limits it now: liquidity (doc 22). With every order capped at the contracts offered within 10 ¢ of fair value on the post-rewards book, the companions fall from median HE 0.67 to 0.18. A typical farm needs 9,000–27,000 contracts against 1,800–5,400 on offer. About 5–10× today's depth restores the fit (0.50 at 5×, 0.66 at 10×). Today the app sizes the order to the book: the Dallas hay farm keeps 90% of the fit on about \$1,200 of cover a month instead of \$6,100. Cover design matters little; a PRF-like linear payout at 90% stays the default. The app now also offers excess-rain cover where it tracks, such as Dallas Jun–Jul (0.58). For each rancher's own policy it shows what two-month contracts would add, for example Apr–May drought cover from 0.33 to 0.51.

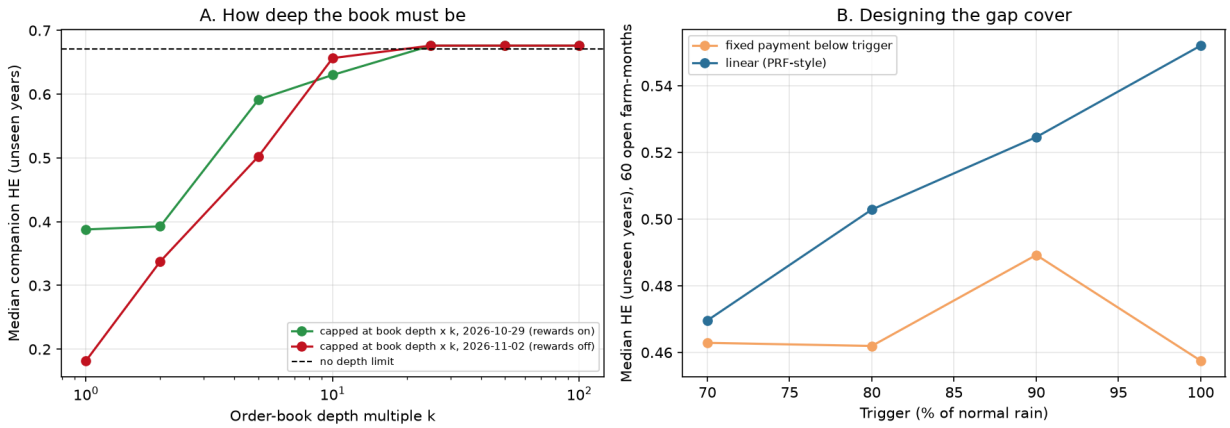


Figure 9: Left: companion fit vs. order-book depth (multiples of today's depth near fair value). Right: gap-cover design, by trigger level and payout shape.

What the rest of the project builds on. The companion is the platform the remaining work expands:

- a liquidity model (market maker or pooled orders) to close the 5–10× depth gap;
- a mock two-month market to demonstrate our contract proposal;
- national coverage for all 38 Kalshi rain stations;
- hedging modeled forage loss instead of the PRF formula;
- temperature risk;
- order placement in Kalshi's demo environment.

5 Synthesis and Next Steps

The platform works, and it gives a farmer an honest answer. That answer is mostly "not yet". With today's contracts, the best hedge near a station removes about a third of PRF's payout variance out of sample, costs about $1.10 per $1 of expected payout against $0.61 for subsidized PRF, and cannot be filled from current order books. Strike binning, the problem Milestone I flagged first, turns out not to matter. The binding problems are where the rain is measured, how long the contract window is, and who is on the other side of the trade. The constructive results are a ranked list of fixes: two-month contracts (no new data needed), seasonal listing, a drought-side market maker, and, as the end point, contracts on a gridded rain index. As our Milestone I update proposed, the tool is most useful as a complement to PRF. The PRF Companion makes that concrete: in the cool-season months PRF leaves open, a Kalshi hedge near a station tracks the farm's rain well on years it never saw (0.61). That is the base the rest of the project builds on.

Reproducibility. Code and data: `https://github.com/Belugaluga2/ECE473-Project` (docs 15–22; demo guide `docs/DEMO.md`). `python scripts/run_optimizer.py`, `run_error_budget.py`, `run_app_test.py -live`, `run_prf_gaps.py`, `run_companion_study.py`, `run_companion_holdout.py`, `run_companion_improve.py`; app: `python scripts/run_app.py` (`?demo` runs the companion). 353 automated tests (20 new for Milestone II).